CyberChef - The Basics

CyberChef is a simple, intuitive web-based application designed to help with various “cyber” operation tasks within your web browser. Think of it as a **Swiss Army knife** for data - like having a toolbox of different tools designed to do a specific task. These tasks range from simple encodings like **XOR** or **Base64** to complex operations like **AES encryption** or **RSA decryption**

Online Access

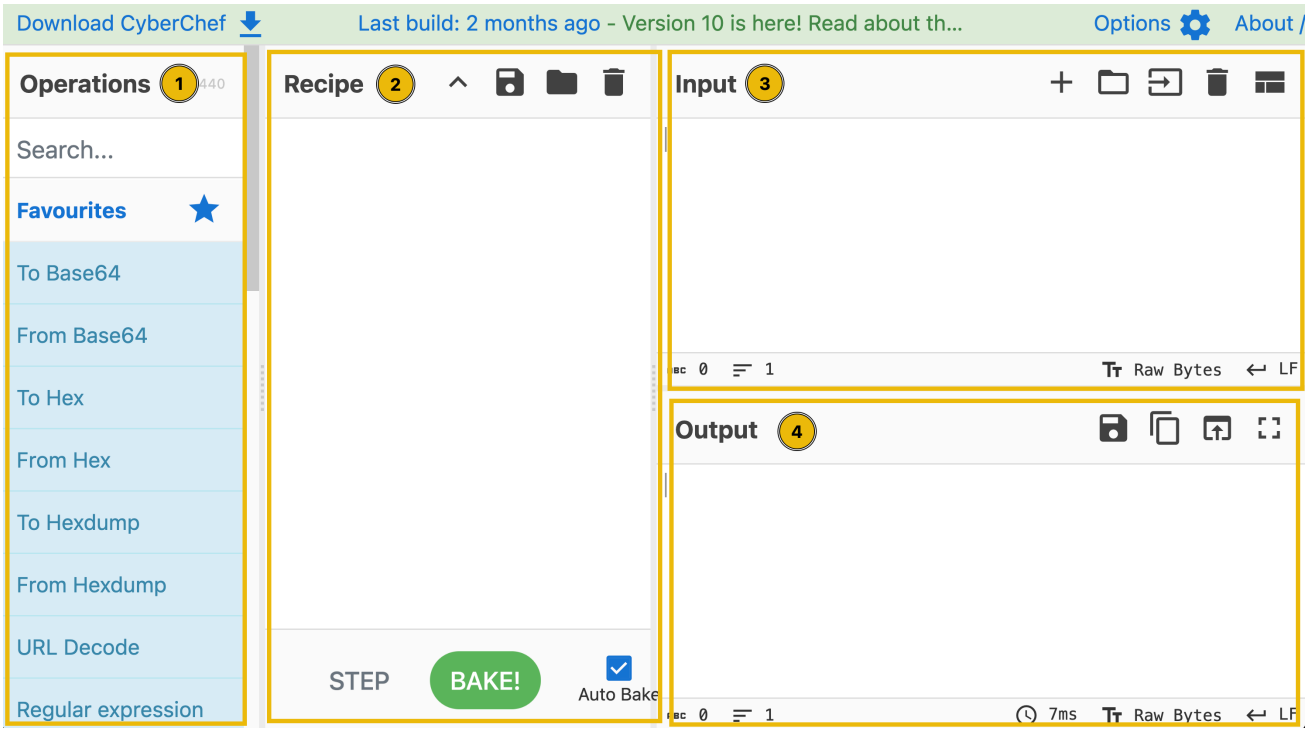
Then, you can click this link to open CyberChef directly within your web browser.

Navigating the interface

CyberChef consists of four areas. Each consists of different components or features.

These are the following areas:

1. Operations
2. Recipe
3. Input
4. Output



The Operations Area

Operations	Description	Examples
From Morse Code	Translates Morse Code into (upper case) alphanumeric characters.	`- .... .-. . .- - ...` becomes `THREATS` when used with default parameters
URL Encode	Encodes problematic characters into percent-encoding, a format supported by URIs/URLs.	`https://tryhackme.com/r/room/cyberchefbasics` becomes `https%3A%2F%2Ftryhackme%2Ecom%2Fr%2Froom%2Fcyberchefbasics` when used with the parameter "Encode all special chars"
To Base64	This operation encodes raw data into an ASCII Base64 string.	`This is fun!` becomes `VGhpcyBpcyBmdW4h`
To Hex	Converts the input string to hexadecimal bytes separated by the specified delimiter.	`This Hex conversion is awesome!` becomes `54 68 69 73 20 48 65 78 20 63 6f 6e 76 65 72 73 69 6f 6e 20 69 73 20 61 77 65 73 6f 6d 65 21`
To Decimal	Converts the input data to an ordinal integer array.	`This Decimal conversion is awesome!` becomes `84 104 105 115 32 68 101 99 105 109 97 108 32 99 111 110 118 101 114 115 105 111 110 32 105 115 32 97 119 101 115 111 109 101 33`
ROT13	A simple Caesar substitution cipher which rotates alphabet characters by the specified amount (default 13).	`Digital Forensics and Incident Response` becomes `Qvtvgny Sberafvpf naq Vapvqrag Erfcbafr`

The Recipe Area

select, arrange, and fine-tune operations to suit your needs. This is where you take control, defining each operation's arguments and options precisely and purposefully.

Features include the following:

- `Save recipe` : This feature allows the user to save selected operations.
- `Load recipe` : Allows the user to load previously saved recipes.
- `Clear Recipe` : This feature will enable users to clear the chosen recipe during usage.

Input Area

he input area provides a user-friendly space where you can easily input text or files by pasting, typing, or dragging them to perform operations.

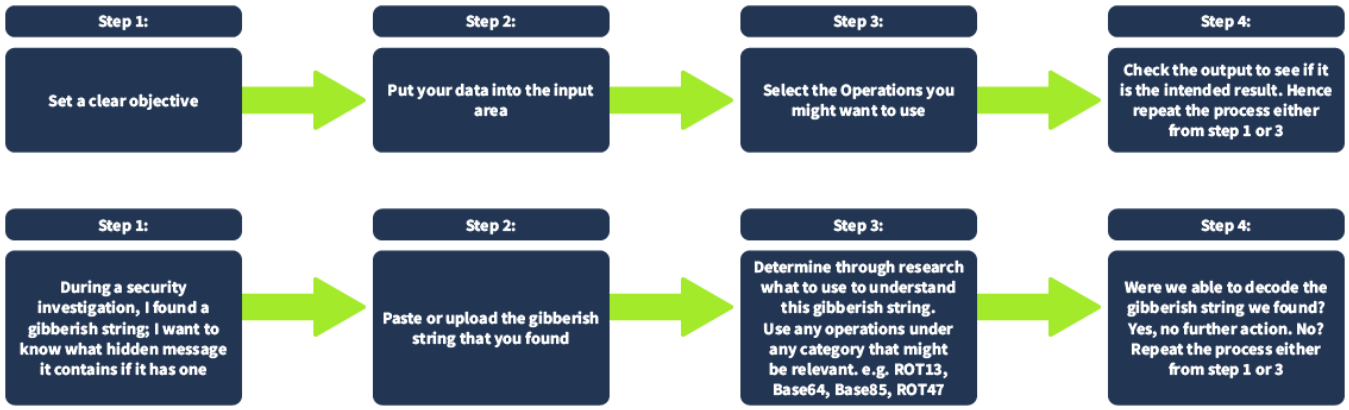
- `Add a new input tab` : This is where an additional tab is created for the user to use different values from the previous tab.
- `Open folder as input` : This feature allows users to upload a whole folder as input value.
- `Open file as input` : This feature allows the user to upload a file as its input value.
- `Clear input and output` : This feature allows the user to clear any input values inserted and the corresponding output value.
- `Reset pane layout` : This feature brings the tool's interface to its default window sizes.

Output Area

The output area is a visual space that showcases the data processing results. It neatly presents the outcomes of any manipulations or transformations you have applied to the input data,

allowing for a clear and intuitive display of the processed information.

- `Save output to file`: This feature allows the users to save the result into a .dat file.
- `Copy raw output to the clipboard`: This feature allows users to copy raw output directly to their clipboard, allowing them to quickly copy the results for use in other applications or documents.
- `Replace input with output`: This feature allows users to quickly overwrite the input data based on the operations' results.
- `Maximise output pane`: This feature brings the tool's interface to its default window sizes.



Practice, Practice, Practice

Extractors

Specific	Description
Extract IP addresses	Extracts all IPv4 and IPv6 addresses.
Extract URLs	Extracts Uniform Resource Locators (URLs) from the input. The protocol (HTTP, FTP, etc.) is required, otherwise there will be far too many false positives.
Extract email addresses	Extracts all email addresses from the input.

Date and Time

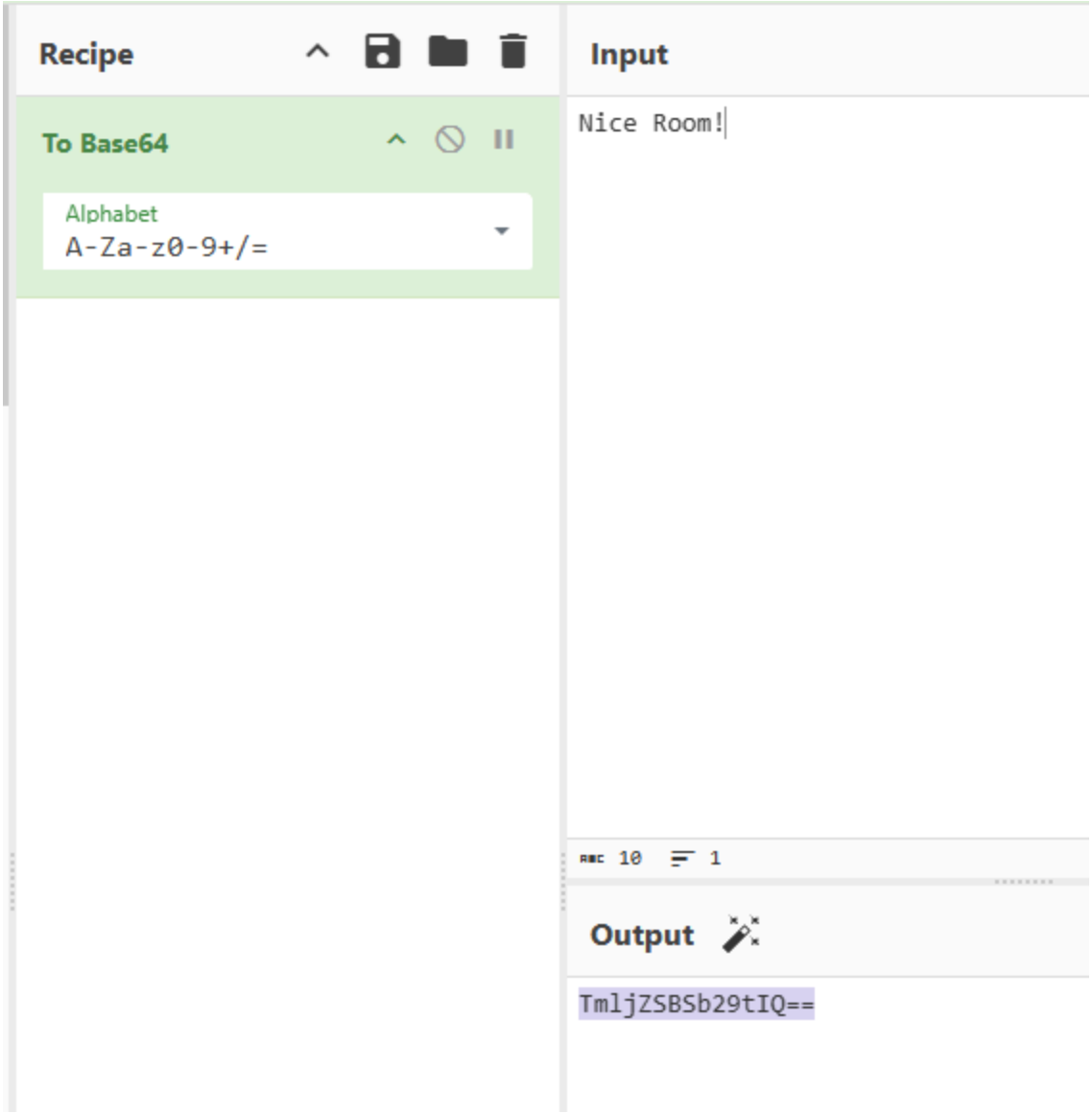
Specific	Description
From UNIX Timestamp	Converts a UNIX timestamp to a datetime string.
To UNIX Timestamp	Parses a datetime string in UTC and returns the corresponding UNIX timestamp.

Data Format

Operations	Description	Examples
From Base64	This operation decodes data from an ASCII Base64 string back into its raw format.	`V2VsY29tZSB0byB0cnloYWNrbWUh` becomes `Welcome to tryhackme!`
URL Decode	Converts URI/URL percent-encoded characters back to their raw values.	`https%3A%2F%2Fgchq%2Egithub%2Eio%2FCyberChef%2F` becomes `https://gchq.github.io/CyberChef/`
From Base85	notation for encoding arbitrary byte data. It is usually more efficient than Base64. This operation decodes data from an ASCII string (with an alphabet of your choosing, presets included).	`BOu!rDlj7BEbo7` becomes `hello world`
From Base58	is a notation for encoding arbitrary byte data. It differs from Base64 by removing efficiently misread characters (i.e. l, I, 0, and O) to improve human readability.	`AXLU7qR` becomes `Thm58`
To Base62	is a notation for encoding arbitrary byte data using a restricted set of symbols that humans can conveniently use and process by computers. The high number base results in shorter strings than with the decimal or hexadecimal system	`Thm62` becomes `6NiRkOY`

Practical

What is the base64 encoded value of the string "**Nice Room!**"?



What is the URL decoded value for `https%3A%2F%2Ftryhackme%2Ecom%2Fr%2Froom%2Fcyberchefbasics` ?

Recipe

URL Decode

- [x] Treat "+" as space

Input

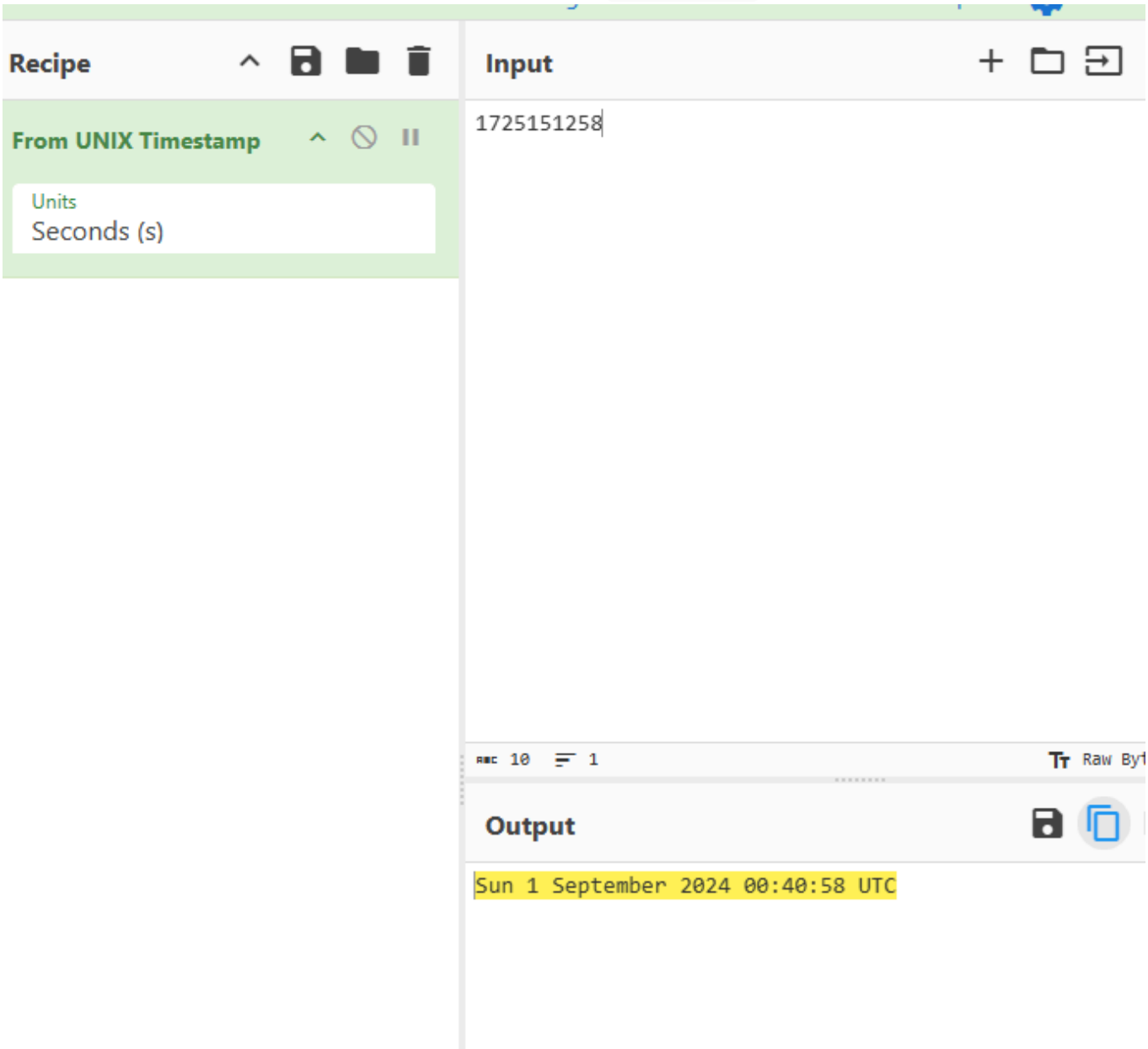
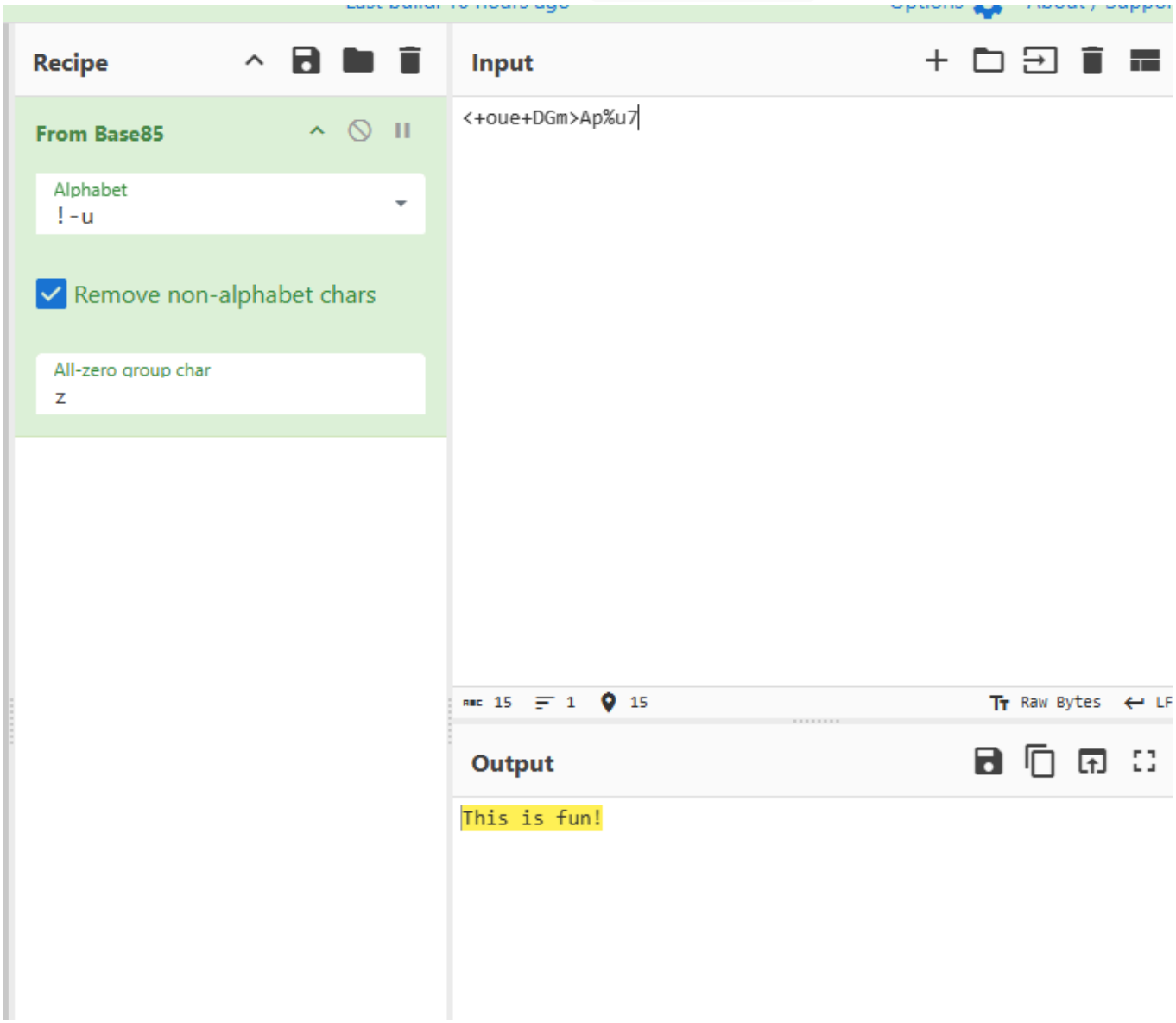
```
https%3A%2F%2Ftryhackme%2Ecom%2Fr%2Froom%2Fcyberchefbasics
```

58 1 58 Raw Bytes LF

Output

```
https://tryhackme.com/r/room/cyberchefbasics
```

What is the datetime string for the Unix timestamp `1725151258`



What is the Base85 decoded string of the value `<+oue+DGm>Ap%u7`